

Justin T. Becker

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EDUCATION

University of Washington

Master of Science in Computer Science; GPA: 3.96

Coursework: Formal Verification, Robotics

Seattle, WA

June 2026

University of Maryland

Bachelor of Science in Computer Engineering; GPA: 3.33

College Park, MD

May 2020

EXPERIENCE

Microsoft

Software Engineer II

Redmond, WA

August 2020 - July 2026

- Developer on the Prism emulator team, enabling dynamic recompilation of x86 and x64 binaries on ARM64 processors, including feature work, as well as retrofitting legacy components to support cross-architecture scenarios
- Redesigned key code paths in Prism, including handrolled assembly, improving game performance by up to 50% in some cases, and up to 2x faster in some user scenarios.
- Performed performance analysis of generated code using WPA, and made substantial changes to code generation resulting in 40% performance improvements in some benchmarks.
- Wrote a custom fuzzer framework to find and fix bugs in our emulation, generating over 100 trillion test cases resulting in the discovery of a number of pernicious bugs.
- Black-box debugged 3rd party applications to root cause and solve compatibility issues under emulation, including AAA games with anti-tamper protection, and other software with anti-debugging measures
- Collaborated with 3rd parties to improve compatibility of their applications, including kernel mode anti-cheat packages.
- Improved support in the Windows NT kernel for emulation scenarios, as well as general improvements for ARM64 machines.
- Analyzed assembly listings in Ghidra for ARM64, ARM32, X86, and X64 binaries for both Windows and Linux
- Developed ARM32 and ARM64 emulators for Linux hosts, including support for syscall translation
- Identified and reported bugs in the MSVC tool-chain, with specific emphasis on ARM64EC and ARM64X code generation
- built and maintained CI pipelines for my team using Azure DevOps Pipelines preventing behavioral regressions
- Enabled new ARM64X “chameleon binary” technology across the Windows code base, allowing ~500 more Windows system DLLs to be loaded in both x64 and ARM64 processes, much of this work being accomplished by bespoke automation tooling I developed

Microstrategy

Software Engineering Intern

Tysons, VA

May 2019 - August 2019

- Built native Business Intelligence Android application in Java and Kotlin, for AndroidTV using Leanback
- Integrated Bluetooth Low Energy based device discovery, to allow for presence based workflows

L3Harris Applied Defense Solutions

Software Engineering Intern

Columbia, MD

May 2018 - January 2019

- Collaborated with Aerospace Engineers to build a Space Mission Planning and Simulation tool using C# and F#
- Supported, maintained, and developed for a 24/7 production data processing system using Java and Kotlin
- Designed monitoring system using InfluxDB to alert operators in realtime of outages and performance degradation

Lockheed Martin, Rotary and Mission Systems

Software Engineering Intern

Moorestown, NJ

May 2017 - January 2018

- Collaborated with systems engineers to develop mission critical software in Java for naval weapons systems

Web Development Intern

June 2016 - May 2017

- Re-architected a legacy web service using modern web technologies such as AngularJS, Java Spring, and MongoDB

PROJECTS

LANBucket

2026

- Built tool for high performance, distributed, file sharing over a local network requiring zero configuration from users, using C++ and WinUI 3.

Terrapin Rockets

September 2016 - May 2019

- Built real-time control software to guide a faux space lander during its mission using an Arduino

Robotics@Maryland

September 2016 - April 2017

- Researched and implemented a computer vision system, using OpenCV, to perform classification tasks

LEADERSHIP

Bitcamp

Executive Director

College Park, MD

May 2019 - May 2020

- Led the 75+ person organizing team of Bitcamp, the 3rd largest college hackathon in the US
- Directly managed a team of 10 directors, spanning Technology, Logistics, Marketing, Sponsorship, and Experience

Director of Logistics

August 2018 - May 2019

- Led and managed a cross disciplinary student team of 22 organizers, planning food, power, networking, A/V, scheduling, volunteers, mentors, travel, highschool outreach, hardware, and workshops for 1,300 participants

SKILLS

Languages: C/C++, Assembly, Java, Python, Bash, Powershell | MATLAB, C#, JS | Kotlin, Rust, Haskell

Tools: Ghidra, WinDbg, GDB, SQL | Docker, MongoDB, InfluxDB, ROS | OpenCV, Angular

Software: Unix, Git, Visual Studio, Android Studio | Jira, Azure DevOps, IntelliJ | Xilinx, ClearCase

HONORS AND AWARDS

Dean's List

Spring 2016, Fall 2019, Spring 2020

Eagle Scout Rank

May 2016